Lift Right — Requirements Specification

CET257 Enterprise Project — Assignment 1, Task A1 — Team Speed

1.Purpose and Scope

This document sets out the functional and non-functional requirements for the Lift Right prototype as built and demonstrated by Team Speed. Requirements are prioritised using MoSCoW (Must / Should / Could) and each is marked with a Status reflecting whether it is implemented in the current build, so the document is an honest record of what has actually been delivered rather than an aspirational feature list. Where a requirement changed following client feedback, this is noted against the relevant meeting in the A4 Meeting Minutes.

2. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement	Priority	Status
FR1	The system shall display a first-person 360° warehouse scene that the worker can freely look around using the mouse.	Must	Implemented
FR2	The system shall allow the worker to move between the Loading Dock, Warehouse Floor and Racking Aisle zones using W-A-S-D controls.	Must	Implemented
FR3	The system shall require the worker to create an account or log in before starting a training attempt.	Must	Implemented
FR4	The system shall present six sequential manual-handling checkpoints, each covering one stage of the safe lift procedure.	Must	Implemented
FR5	The system shall only offer load-handling methods (Manual Lift, Team Lift, Trolley/Equipment) that are appropriate for the weight of the load presented.	Must	Implemented
FR6	The system shall present a knowledge-check quiz question after each checkpoint and record whether the answer was correct.	Must	Implemented
FR7	The system shall award points for correct answers, up to a maximum of 100 points per run, and apply a 15-point penalty per missed checkpoint.	Must	Implemented
FR8	The system shall trigger a bonus ‘Golden Crate	Should	Implemented

	Drill’ event after three consecutive correct answers.		
FR9	The system shall trigger a ‘Hazard-Spotting Stand-Down’ recovery event after three consecutive missed answers.	Should	Implemented
FR10	The system shall display a live HUD showing a countdown timer, current score and current zone throughout the simulation.	Must	Implemented
FR11	The system shall record each attempt's score, time-on-task and login activity against the worker's account.	Must	Implemented
FR12	The system shall provide an Admin Command Center where a supervisor can view per-worker attempts, best score and login history.	Must	Implemented
FR13	The system shall allow the supervisor to export the full training record to CSV in one click.	Should	Implemented
FR14	The system shall allow a worker to replay a completed run from the review screen without losing their previous best score.	Could	Implemented

3. Non-Functional Requirements

Non-functional requirements describe the quality attributes the system must meet: performance, usability and compatibility. The Evidence column records how each was checked.

ID	Category	Requirement	Evidence
NFR1	Performance	The 360° scene shall load and become interactive within 3 seconds on a standard office laptop.	Timed manually on team laptops during Week 2–Week 3 build.
NFR2	Compatibility	The application shall run entirely in a modern desktop browser (Chrome, Edge, Firefox) with no install or plugin required.	Cross-browser smoke test before each client demo.
NFR3	Usability	The interface shall remain legible and usable at both 1920x1080 and 1366x768 screen resolutions.	Checked against both resolutions during HUD polish (Week 7).

NFR4	Security	Worker account passwords shall not be displayed in plain text in the interface.	Confirmed by code review of the login/account screens.
NFR5	Performance	The HUD shall update score, timer and zone information with no perceptible lag (<200ms) during gameplay.	Observed during playtesting sessions, Weeks 5–7.
NFR6	Deployability	The application shall be a single self-contained file requiring no server-side installation for the client to trial it.	Verified by running the build from a fresh machine with no prior setup.
NFR7	Reliability	The scoring and event logic shall run entirely client-side so the demo does not depend on network connectivity.	Verified by running a full attempt offline.
NFR8	Usability	The system shall retain a worker's stats between sessions on the same browser/device.	Verified by logging out and back in mid-testing.
NFR9	Performance	The admin CSV export shall complete in under 2 seconds for up to 100 worker records.	Timed against a synthetic 100-record test set.
NFR10	Usability	The colour palette and HUD contrast shall remain readable under typical office lighting conditions.	Checked visually during rehearsal (Week 7) under normal room lighting.

4. Requirements Traceability

All Must and Should requirements above trace to either the PACT Analysis (see the People, Activities and Contexts findings) or to feedback recorded in the Week 3 and Week 5 client meetings in the A4 Meeting Minutes. FR14 (Could) was added after the Week 5 client meeting as a small, low-cost scope addition rather than being part of the original brief.

5. Assumptions

- End users have access to a reasonably modern desktop or laptop browser; no support for outdated or unsupported browsers is assumed or required.
- For the prototype stage, client-side storage of accounts and attempts is an acceptable substitute for a full backend; this is flagged as a future enhancement rather than a permanent design choice.
- Workers attempt the simulation on a shared or personal desktop/laptop rather than a mobile device; no mobile-specific requirements are included in this version.